

Propagation of a Vertical Hydraulic Fracture

R. P. NORDGREN

SHELL DEVELOPMENT CO.
HOUSTON, TEX.

ABSTRACT

This paper treats the propagation of hydraulic fractures of limited vertical extent and elliptic cross-section with the effect of fluid loss included. Numerical and asymptotic approximate solutions in dimensionless form give the fracture length and width at any value of time for any set of physical parameters. The insight provided by the dimensionless results and approximate solutions should be useful in the design of fracture treatments.

INTRODUCTION

The theory and practice of hydraulic fracturing has been reviewed by Howard and Fast.¹ Therefore, we confine our discussion of previous investigations to those pertinent to the present study of the propagation of vertical fractures.

An important theoretical result is Carter's formula² for the area of a fracture of constant width formed by injection at constant rate with fluid lost to the formation. For a vertical fracture of constant height, Carter's formula gives fracture length as a function of time. In general, Carter's assumption of constant width is not realistic. However, at large values of time the effect of this assumption becomes insignificant since the effect of fluid loss dominates.

The width of a vertical fracture was first investigated by Khristianovic and Zheltov³ under the assumption that the width does not vary in the vertical direction. Thus, a state of plane strain prevails in horizontal planes and the width can be determined as the solution of a plane elasticity problem. An approximate solution is found in Ref. 3 upon neglect of fluid loss, fracture volume change, and pressure variation along the fracture. The fracture length is determined by the condition of finite stress at the fracture tip. Baron *et al.*⁴ and Geertsma and de Klerk⁵ have included the effect of fluid loss in the approach of Ref. 3. Geertsma and de Klerk⁵ give simple approximate formulas for

fracture length and width.

A different approach to the determination of fracture width was taken by Perkins and Kern.⁶ They considered a vertically limited fracture under the assumption of plane strain in vertical planes perpendicular to the fracture plane. The cross-section of the fracture is found to be elliptical, and the maximum width decreases along the fracture according to a simple formula that contains the fracture length. In the derivation of this formula,⁶ fluid loss and fracture volume change are neglected in the continuity equation and the fracture length is not determined. In a subsequent application,⁷ a "reasonable" fracture length was assumed. Carter's formula² for length and the width formula of Perkins and Kern⁶ are both cited by Howard and Fast,¹ and combined use of the two formulas is believed to be common practice.

The present theoretical investigation is concerned with vertically limited fractures of the type studied by Perkins and Kern.⁶ However, we include the effects of fluid loss and fracture volume change in the continuity equation. Consequently, fracture length is determined as part of the solution. General results for the variation of fracture width and length with time are obtained in dimensionless form by a numerical method. In addition, asymptotic solutions are derived for large and small values of time. The small-time solution is also the exact solution for the case of no fluid loss to the formation. For large values of time our asymptotic formula for fracture length is identical with Carter's formula² at large time. Our large-time formula for fracture width differs from the formula of Perkins and Kern⁶ by a numerical factor that varies along the fracture. In comparison with our formula, this formula⁶ overestimates the width by 12 percent at the well and 24 percent at the midlength of the fracture. At early times Perkins and Kern's formula⁶ for width in terms of length is again a fair approximation to our result. However, our formula for length differs from Carter's formula,² which is not applicable since the neglected width variation is important at early times.

The results for the width of a vertically limited fracture as obtained here and in Ref. 6 differ from the results for vertically constant fractures.³⁻⁵ In particular, our large-time solution for width differs from that of Geertsma and de Klerk⁵ by a factor proportional to the fourth root of fracture height

Original manuscript received in Society of Petroleum Engineers office June 1, 1970. Revised manuscript received March 20, 1972. Paper (SPE 3009) was presented at SPE 45th Annual Fall Meeting, held in Houston, Oct. 4-7, 1970. © Copyright 1972 American Institute of Mining, Metallurgical, and Petroleum Engineers, Inc.

¹References given at end of paper.

This paper will be printed in *Transactions* volume 253, which will cover 1972.

divided by fracture length. This difference may be significant in practical applications, and we will discuss it further.

In the next section we formulate the problem of vertically limited fractures along the lines of Ref. 6. The underlying assumptions are discussed in some detail. The basic equations then are cast in dimensionless form suitable for numerical solution, which is carried out in Appendix A. The last section of the paper gives a comparison with previous investigations and a discussion of the applicability of the present results.

FORMULATION

We consider a vertical fracture propagating in a straight line away from the well (Fig. 1). As in Ref. 6, the vertical height of the fracture is assumed to be limited to a constant distance, h , by layers of fracture resistant rock. The fracture resistance may be due to higher horizontal tectonic stress or higher fracture strength in the rock surrounding the reservoir.

Let x, y, z be a system of rectangular Cartesian coordinates with x axis in the direction of fracture propagation, z axis parallel to the well axis, and origin at the well face (Fig. 1). Thus, the fracture lies in the x - z plane with $0 \leq x \leq L$ and $|z| \leq \frac{1}{2}h$.

Following Ref. 6, we assume that an isotropic homogeneous elastic material surrounds the fracture. The homogeneity assumption may not be fully consistent with our earlier assumption that fracture resistant rock bounds the fractured reservoir. However, on the basis of St. Venant's principle in the theory of elasticity, the elastic behavior near the fracture is controlled mainly by the reservoir rock and moderately different surrounding rock should have a small effect on fracture width. We confine attention to the stage of fracture propagation when the length, L , is much greater than the height, h , and all quantities are expected to vary slowly with x along most of the fracture. This assumption will be verified after obtaining the

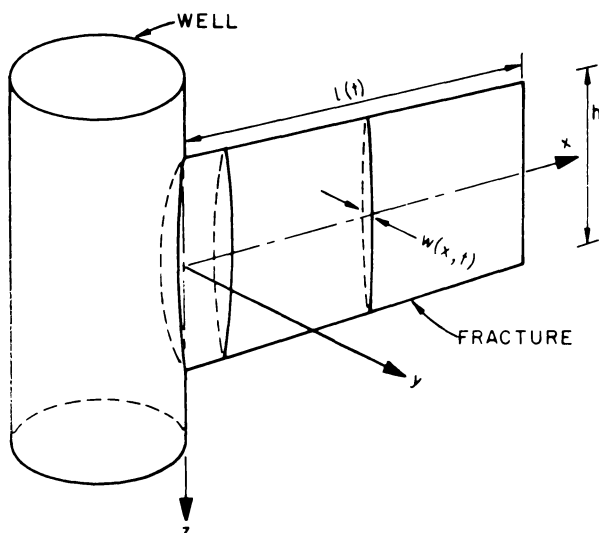


FIG. 1 — FRACTURE GEOMETRY.

solution. In view of the expected slow variations with x , an approximate state of plane strain prevails in vertical planes perpendicular to the fracture (planes $x = \text{const.}$).

Before fracturing, a compressive stress (S) acts on planes parallel to the fracture in the reservoir. Different values for S in the rock surrounding the reservoir again should have a small effect. The fluid in the fracture is under a pressure $S + \Delta p$, and we neglect variation of Δp with z so that $\Delta p = \Delta p(x, t)$. Then, by the plane-strain solution for a crack under constant pressure,⁸ the fracture is elliptical and the width, $w(x, t)$, is given by

$$w = \frac{(1 - \nu)}{G} (h^2 - 4z^2)^{1/2} \Delta p, \quad |z| \leq \frac{1}{2}h$$

$$w = 0, \quad |z| \geq \frac{1}{2}h \quad \dots \dots (1)$$

where G and ν are the bulk shear modulus and Poisson's ratio for the formation. The plane-strain solution neglects the effect of pore-pressure variation on elastic response which is a small effect in fracture problems.⁹

Next, we consider the flow of fluid in the fracture. The continuity equation for flow of an incompressible fluid in the fracture can be written as

$$\frac{\partial q}{\partial x} + q_\ell + \frac{\partial A}{\partial t} = 0, \quad \dots \dots (2)$$

where $q(x, t)$ is the volume rate of flow through a cross-section ($x = \text{const.}$) of the fracture, $q_\ell(x, t)$ is the volume rate of fluid loss to formation per unit length of fracture, and $A(x, t)$ is the cross-sectional area of the fracture.

We assume that q is related to the pressure gradient by the classical solution¹⁰ for laminar flow of a Newtonian viscous fluid in an elliptical tube of semiaxes $\frac{1}{2}h$ and $\frac{1}{2}w_{\text{max}}$ with $h \gg w_{\text{max}}$, namely,

$$q = - \frac{\pi W^3 h}{64\mu} \frac{\partial \Delta p}{\partial x}, \quad W \equiv w_{\text{max}}, \quad \dots (3)$$

where μ is the viscosity of the fracturing fluid. The range of validity for laminar flow has been discussed in Ref. 6. By Eq. 1,

$$W = w|_{z=0} = \frac{(1 - \nu)h}{G} \Delta p \quad \dots \dots (4)$$

and Eq. 3 becomes

$$q = - \frac{\pi G}{256(1 - \nu)\mu} \frac{\partial}{\partial x} (W^4) \quad \dots \dots (5)$$

The cross-sectional area of the elliptical fracture is given by

$$A = \int_{-h/2}^{h/2} w dz = \frac{\pi}{4} Wh \quad (6)$$

The fluid-loss rate, q_ℓ , is related to the loss rate, u_ℓ , per unit area of fracture surface by

$$q_\ell = 2 \int_{-h/2}^{h/2} u_\ell dz \quad (7)$$

and if u_ℓ is independent of z , then

$$q_\ell = 2hu_\ell \quad (8)$$

Experiments^{1,11} suggest that for many fracturing fluids the loss rate can be taken as

$$u_\ell = \frac{C}{\sqrt{t - \tau}} \quad (9)$$

where C is the fluid-loss coefficient and τ is the time at which flow begins. Coefficient C depends on various fluid and formation parameters, and it is usually determined experimentally. For some fluids, Eq. 9 should be modified to include the so-called "spurt loss" at $t = \tau$. The effect of spurt loss on fracture length is found to be small in general (Appendix E). We adopt Eq. 9 in the present study, although the conditions under which it was verified experimentally are not strictly satisfied. Namely, the pressure difference between fracture pressure $S + \Delta p$ and formation pore pressure p_f , i.e.,

$$S + \Delta p - p_f \quad (10)$$

contains Δp , which varies with x and t . In some cases, $\Delta p \ll S - p_f$ so that the assumption of constant pressure differences should lead to meaningful results. Furthermore, we will find that Δp varies slowly with t and x and use of an average value for Δp in Eq. 10 may be acceptable. The condition of one-dimensional flow is not strictly satisfied either, although in view of the slow variation of Δp with x , the resulting error should be small. Theoretical refinements in the determination of q_ℓ would require additional experimental research on filter cake behavior under time-dependent pressure. Also, consideration of two-phase flow would be necessary if the fracturing fluid differs significantly from the pore fluid.

At the well ($x=0$) a boundary condition involving q and p must be specified. This condition will depend on the $q(p)$ relation for pumping equipment used in the fracturing process. Here we treat the case of constant inflow rate

$$q(0, t) = q_1 = \text{const.} \quad (11)$$

Other boundary conditions can be treated by the numerical method given in Appendix A.

Initially, the fracture is closed, i.e.,

$$W(x, 0) = 0 \quad (12)$$

Furthermore, the fracture remains closed for $x \geq L(t)$, i.e.,

$$W(x, t) = 0, \quad x \geq L(t) \quad (13)$$

where the fracture length $L(t)$ is to be determined as part of the solution.

Upon the substitution of Eqs. 5, 6, 8 and 9 into Eq. 2, we obtain

$$\frac{G}{64(1 - \nu)\mu h} \frac{\partial^2}{\partial x^2} (W^4) = \frac{8C}{\pi\sqrt{t - \tau(x)}} + \frac{\partial W}{\partial t} \quad (14)$$

$$0 \leq x \leq L, \quad t > 0$$

where $\tau(x)$ is the time at which the fracture opened at x , i.e.,

$$\tau[L(t')] = t', \quad 0 \leq t' \leq t \quad (15)$$

In other words, $\tau(x)$ is the function inverse to $L(t)$. Substitution of Eq. 5 into Eq. 11 gives

$$-\left[\frac{\partial}{\partial x} (W^4) \right]_{x=0} = \frac{256(1 - \nu)\mu}{\pi G} q_1 \quad (16)$$

Now our problem is the solution of the nonlinear partial differential equation (Eq. 14) for $W(x, t)$ subject to initial condition (Eq. 12) and boundary conditions (Eqs. 13 and 16) with $L(t)$ also to be determined.

To minimize the amount of computation, it is convenient to cast the governing equations into dimensionless form. Thus, if we set

$$x = ax_0, \quad L = aL_0$$

$$t = Bt_0, \quad W = eW_0$$

where

$$a = \pi \left[\frac{(1 - \nu)\mu q_1^5}{256C^5 Gh^4} \right]^{1/3}$$

$$B = \pi^2 \left[\frac{(1 - \nu)\mu q_1^2}{32C^5 hG} \right]^{2/3}$$

$$e = \left[\frac{16(1 - \nu)\mu q_1^2}{C^2 Gh} \right]^{1/3} \quad (17)$$

then Eqs. 12 and 16 become

$$\frac{\partial^2}{\partial x_D^2} (W_D^4) = \frac{1}{\sqrt{t_D - \tau_D(x)}} + \frac{\partial W_D}{\partial t_D}$$

$$-\frac{\partial}{\partial x_D} (W_D^4) \Big|_{x_D=0} = 1, \quad W(x_D, 0) = 0$$

$$W_D(x_D, t_D) = 0 \quad \text{for } x_D > L_D,$$

$$\tau_D[L_D(t_D')] = t_D' \quad \text{for } 0 \leq t_D' \leq t_D.$$

..... (18)

Eqs. 18 have been solved for $W_D(x_D, t_D)$ by the numerical method of Appendix A.

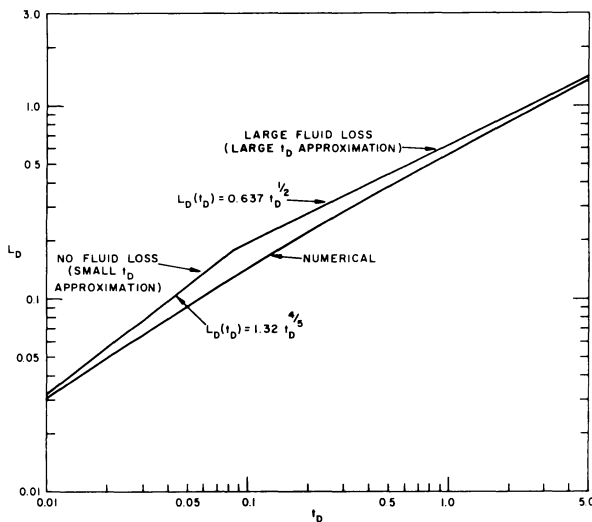


FIG. 2 — DIMENSIONLESS FRACTURE LENGTH VS DIMENSIONLESS TIME.

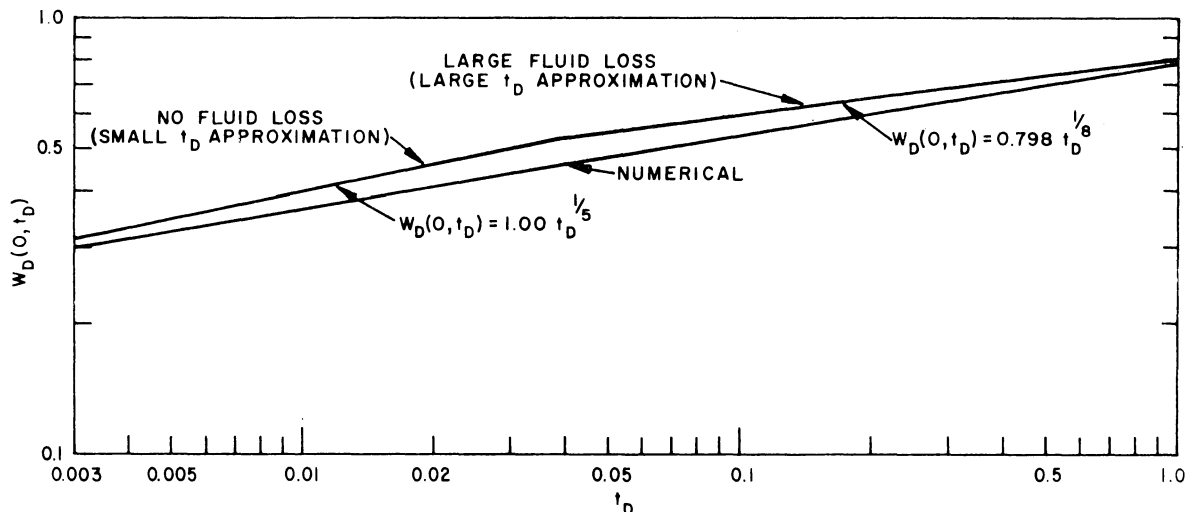


FIG. 3 — DIMENSIONLESS MAXIMUM FRACTURE WIDTH AT WELL VS DIMENSIONLESS TIME.

DISCUSSION OF RESULTS

Results have been obtained by three different methods. The numerical method of Appendix A has been applied to the dimensionless equation (Eq. 18). The fracture length $L_D(t_D)$ and the fracture width at the well $W(0, t_D)$ vs time t_D are shown in Figs. 2 and 3.

An approximate solution in elementary functions is derived in Appendix B upon neglect of the term $\partial A / \partial t$ in Eq. 2. This approximation is valid for sufficiently large time as indicated in Appendix B and confirmed by comparison with the numerical results in Figs. 2 and 3. For an error in the approximation of less than 10 percent $t_D > 1.0$.

The case of no-fluid loss out of the fracture is treated in Appendix C. A similarity transformation in combination with the numerical method of Appendix A furnishes a solution for any value of time. The results of Appendix C also constitute an approximate solution for the case of fluid loss out of the fracture at sufficiently small time. By comparison with numerical results in Figs. 2 and 3, for an error in the approximation of less than 10 percent, $t_D < 0.01$. Also, t_D must be large enough that $L \gg h$, as discussed below.

Fig. 4 shows the variation of maximum fracture width along its length, x , in dimensionless form for the approximate solutions of Appendices B and C. These two width variations are remarkably similar. Widths obtained by the numerical method lie close to the width curves in Fig. 4 and are not shown.

For any particular problem the fracture length and width at any time can be obtained from Figs. 2 through 4 by means of Eq. 17. One merely computes the value of t_D by Eq. 17, reads $L_D(t_D)$ and $W_D(0, t_D)$ from Figs. 2 and 3, and computes $L(t)$ and $W(0, t)$ from Eq. 17. Then, the approximate variation of maximum fracture width with distance from the well follows from Fig. 4. Also, the pressure Δp is obtained from W by Eq. 4.

The approximate solutions of Appendices B and C provide considerable insight into the effect of

physical parameters on fracture length, width and pressure. For the case of large fluid-loss rate or large time (Appendix B), the fracture length and the maximum width at the borehole are given by Eqs. B-5 and B-9 as

$$L(t) = \frac{q_i t^{1/2}}{\pi Ch}$$

$$W(0, t) = 4 \left[\frac{2(1 - \nu)\mu q_i^2}{\pi^3 GCh} \right]^{1/4} t^{1/8} \quad (19)$$

For the case of no-fluid loss, or small time, (Appendix C) these same quantities are

$$L(t) = 0.68 \left[\frac{Gq_i^3}{(1 - \nu)\mu h^4} \right]^{1/5} t^{4/5}$$

$$W(0, t) = 2.5 \left[\frac{(1 - \nu)\mu q_i^2}{Gh} \right]^{1/5} t^{1/5} \quad (20)$$

As seen from Eqs. 19 and 20, the fracture length and width grow faster with time in the no-loss (small-time) case than in the large-loss (large-time) case. The manner in which other parameters affect fracture size can also be seen from Eqs. 19 and 20. For example, L is proportional to q_i in Eq. 19 and $q_i^{3/5}$ in Eq. 20. In Eq. 19, L is not affected by G or μ .

Comparison is made with the approximate solution of Perkins and Kern⁶ in Appendix D. It is found that, if Eqs. 19 and 20 are cast in the form $W(0, t)$ as a function of L (by elimination of t), then the approximate solution⁶ is fairly accurate. However, at early times Eq. 20 for $L(t)$ differs from Carter's formula,² which was based on the assumption of constant width. At large times, Eq. 19 for $L(t)$ agrees with the large-time form of

Carter's formula,² since the fluid loss effect now is dominant.

Detailed studies of effects of parameter variations can be made through Eq. 17 and the dimensionless results in Figs. 2 and 3 which should be useful in the design of fracture treatments. For example, the fluid-loss coefficient, C , required to achieve a desired fracture length after a specified pumping time can be determined from these results. Pumping pressures required to achieve a desired injection rate can be determined. Also, the fracture treatment can be designed so that the fracture opening admits a certain size of proppant.

When applying the foregoing results, the basic assumptions of the formulation must be kept in mind. Probably the most important assumptions are that fracture height is limited vertically and that fracture length is much greater than height ($L \gg b$). As seen from Fig. 4, the latter assumption assures slow variation of width, w , with x which validates the basic assumption of plane strain. If the condition $L \gg b$, in particular immediately after fracture initiation, is not fulfilled or if the fracture is not vertically limited, then other theoretical models of fracture propagation are more appropriate. For a vertical fracture with $L \ll b$ and injection over all of the height, the horizontal plane strain model of Refs. 3 through 5 appears to be more appropriate. If injection is limited to a short interval, then a radially propagating vertical fracture should be a suitable model as suggested by Perkins and Kern.⁶ This model has been treated approximately by Geertsma and de Klerk⁵ and others. There are significant differences between these solutions³⁻⁵ and the present one. In particular, for a linearly propagating (one-sided) vertical fracture, the approximate fracture width at the well as given by Geertsma and de Klerk (for $\nu = 0.25$) as

$$W(0) = 2.1 \left(\frac{\mu q_i L^2}{2Gh} \right)^{1/4},$$

which differs from our large-time solution (Eq. 19) by the factor $0.67(L/b)^{1/4}$. This factor is large when $L/b \ll 1$ and $L/b \gg 1$. As stated above, our solution is not valid for $L/b \ll 1$. On the other hand, we do not believe that the approach of Refs. 3 through 5 is valid for $L/b \gg 1$, since the assumption of vertically precludes fracture closure at the top and bottom. The error thereby induced is roughly proportional to L and, thus, the error becomes large for $L \gg b$. In contrast the error induced in our solution by rapid variations near the fracture tip is roughly proportional to b and this error becomes small for $L \gg b$. For $L \approx b$ a three-dimensional solution is necessary, but lacking at present. Such a solution also would conclusively resolve the ranges of applicability of our results and those of Refs. 3 through 5.

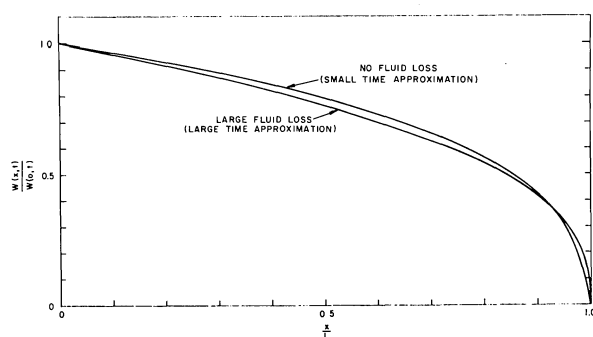


FIG. 4 — VARIATION OF MAXIMUM FRACTURE WIDTH WITH DISTANCE FROM WELL.

NOMENCLATURE

- A = cross-sectional area of the fracture
 C = fluid-loss coefficient
 G = bulk shear modulus of formation
 b = thickness of fractured layer
 K = constant (defined by Eq. C-3)
 L = length of fracture
 L_D = dimensionless L
 p_f = pore pressure in formation
 q = volume rate of flow across a cross-section of the fracture
 q_ℓ = fluid-loss rate per unit length of fracture
 q_i = constant flow rate into fracture at well
 S = normal compressive stress on fracture plane before fracturing
 t = time
 t_D = dimensionless t
 u_ℓ = fluid-loss rate per unit surface area
 v = fluid velocity
 W = fracture width at center of fracture ($w_{\max}$)
 W_D = dimensionless W
 w = fracture width
 x, y, z = rectangular Cartesian coordinates (Fig. 1)
 x_D = dimensionless x
 Δp = pressure in fracture minus S
 μ = viscosity of fracturing fluid
 ν = bulk Poisson's ratio of formation
 $\tau(x)$ = time at which fluid loss begins at distance x from well

REFERENCES

- Howard, G. C. and Fast, C. R.: *Hydraulic Fracturing*, Monograph Series, Society of Petroleum Engineers, Dallas (1970) Vol. II.
- Carter, R. D.: "Derivation of the General Equation for Estimating the Extent of the Fractured Area," Appendix to: "Optimum Fluid Characteristics for Fracture Extension," by G. C. Howard and C. R. Fast, *Drill. and Prod. Prac.*, API (1957) 261-270.
- Khristianovic, S. A. and Zheltov, Y. P.: "Formation of Vertical Fractures by Means of Highly Viscous Liquid," *Proc.*, Fourth World Pet. Cong., Rome (1955) Sec. 2, 579-586.
- Baron, G. *et al.*: "Fracturation hydraulique; bases théoriques, études de laboratoire, essais sur champ," *Proc.*, Seventh World Pet. Cong., Mexico City (1967) Sec. 3, 371.
- Geertsma, J. and de Klerk, F.: "A Rapid Method of Predicting Width and Extent of Hydraulically Induced Fractures," *J. Pet. Tech.* (Dec., 1969) 1571-1581.
- Perkins, T. K. and Kern, L. R.: "Widths of Hydraulic Fractures," *J. Pet. Tech.* (Sept., 1961) 937-949.
- Kern, L. R., Perkins, T. K. and Wyant, R. E.: "Designing Aluminum-Pellet Fracturing Treatments," *Drill. and Prod. Prac.*, API (1961).
- England, A. H. and Green, A. E.: "Some Two-Dimensional Punch and Crack Problems in Classical Elasticity," *Proc.*, Cambridge Phil. Soc. (1963) Vol. 59, 489.
- Geertsma, J.: "Problems of Rock Mechanics in Petroleum Engineering," *Proc.*, First Congress

International Society of Rock Mechanics, Lisbon (1966) 585-594.

- Lamb, H.: *Hydrodynamics*, 6th Ed., Dover Publications, Inc., New York (1932).
- Hall, C. D., Jr. and Dollahide, F. E.: "Performance of Fracturing Fluid Loss Agents Under Dynamic Conditions," *J. Pet. Tech.* (July, 1968) 763-769.
- Ames, W. F.: *Nonlinear Partial Differential Equations in Engineering*, Academic Press, Inc., New York (1965).

APPENDIX A

NUMERICAL SOLUTION

Here we derive a finite-difference analog of Eqs. 12 through 16 and give a numerical method of solution. In order to obtain finite-difference equations which satisfy an over-all fluid balance condition, we begin the derivation with the continuity equation (Eq. 2) together with Eqs. 8, 9 and 11, i.e.,

$$\frac{\partial q}{\partial x} + \frac{2hC}{\sqrt{t - \tau(x)}} + \frac{\partial A}{\partial t} = 0, \quad 0 < x < L \quad (A-1)$$

$$q(0, t) = q_i \quad (A-2)$$

Integration of Eq. A-1 with respect to time from t_m to t_{m+1} with use of the trapezoidal integration formula gives

$$\frac{\Delta t_m}{2} \frac{\partial}{\partial x} (q^{m+1} + q^m) + [4hC\sqrt{t - \tau(x)} + A]_{t_m}^{t_{m+1}} = 0, \quad \dots (A-3)$$

where

$$\Delta t_m = t_{m+1} - t_m, \quad q^m = q(x, t_m)$$

Integration of Eq. A-3 with respect to x from $x_{i-1/2}$ to $x_{i+1/2}$ with use of Eq. 6 and the central value for the integral gives

$$\begin{aligned} & \frac{\Delta t_m}{2 \Delta x} (q_{i+1/2}^{m+1} - q_{i-1/2}^{m+1} + q_{i+1/2}^m - q_{i-1/2}^m) \\ & + 4hC(\sqrt{t_{m+1} - \tau_i} - \sqrt{t_m - \tau_i}) \\ & + \frac{\pi h}{4} (W_i^{m+1} - W_i^m) = 0, \quad \dots (A-4) \end{aligned}$$

where

$$\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}},$$

$$W_i^m = W(x_i, t_m),$$

$$x_i = \frac{1}{2}(x_{i+\frac{1}{2}} + x_{i-\frac{1}{2}}).$$

If we take $x_{1/2} = 0$, then by Eq. A-2,

$$q_{1/2}^m = q_1 \dots \dots \dots (A-5)$$

and by Eq. 2,

$$W_1^0 = 0 \dots \dots \dots (A-6)$$

A finite-difference analog of Eq. 5 is

$$q_{i+\frac{1}{2}} = - \frac{\pi G}{256(1-\nu)\mu \Delta x} [(W_{i+1}^m)^4 - (W_i^m)^4] \dots \dots \dots (A-7)$$

The result of substituting Eq. A-7 in Eq. A-4 is a nonlinear system of equations for W_i^{m+1} with W_i^m considered known. In order to solve this system we write

$$W_i^{m+1} = W_i^m + \Delta W_i^m \dots \dots \dots (A-8)$$

and take

$$(W_i^{m+1})^4 \simeq (W_i^m)^4 + 4(W_i^m)^3 \Delta W_i^m \dots \dots (A-9)$$

On substitution of Eqs. A-6, A-8 and A-9 into Eq. A-4, we obtain a linear tridiagonal system, which is easily solved for ΔW_i^m at $m = 1, 2, \dots$

APPENDIX B

SOLUTION FOR LARGE FLUID-LOSS RATE

Here we assume that the term $\partial A / \partial t$ in Eq. 2 is negligible compared with the fluid-loss rate q_l . Then by Eqs. 8 and 9, Eq. 2 becomes

$$\frac{\partial q}{\partial x} = - \frac{2hC}{\sqrt{t - \tau(x)}}, \dots \dots \dots (B-1)$$

where τ depends on x such that

$$\tau[L(t')] = t', \quad 0 \leq t' \leq t \quad (B-2)$$

Integration of Eq. B-1 with Eq. 11 yields

$$q = q_1 - 2hC \int_0^x \frac{dx}{\sqrt{t - \tau(x)}} \dots (B-3)$$

Since there is no flow out of the end of the fracture [$q(L, t) = 0$], by Eq. B-3

$$\int_0^L \frac{dx}{\sqrt{t - \tau(x)}} = \frac{q_1}{2hC} \dots \dots \dots (B-4)$$

The solution of Eq. B-4 with Eq. B-2 is

$$\tau = \left(\frac{\pi h C x}{q_1} \right)^2, \quad L = \frac{q_1 t^{\frac{1}{2}}}{\pi h C} \dots \dots (B-5)$$

which was obtained previously^{2,5,9} in a somewhat different manner. By Eq. B-5, Eq. B-3 becomes

$$q = q_1 \left[1 - \frac{2}{\pi} \sin^{-1} \zeta \right], \quad \zeta = \frac{x}{L} \dots \dots \dots (B-6)$$

By Eqs. 5 and B-6 and the condition $W(L, t) = 0$, we obtain

$$W = 4 \left[\frac{2(1-\nu)\mu q_1^2}{\pi^3 C G h} \right]^{\frac{1}{4}} t^{\frac{1}{8}} \phi(\zeta) \dots \dots (B-7)$$

where

$$\phi(\zeta) = \left[\frac{W(x, t)}{W(0, t)} \right]_{x=L\zeta} = \left[\zeta \sin^{-1} \zeta + (1 - \zeta^2)^{\frac{1}{2}} - \frac{\pi}{2} \zeta \right]^{\frac{1}{4}} \dots \dots \dots (B-8)$$

The function $\phi(\zeta)$ is shown in Fig. 4. By Eqs. 4, B-5 and B-7, the maximum fracture width and pressure at the origin are given by

$$W(0, t) = 4 \left[\frac{2(1-\nu)\mu q_1^2}{\pi^3 G C h} \right]^{\frac{1}{4}} t^{\frac{1}{8}}$$

$$\Delta p(0, t) = 4 \left[\frac{2G^3 \mu q_1^2}{\pi^3 (1-\nu)^3 C h^5} \right]^{\frac{1}{4}} t^{\frac{1}{8}} \dots (B-9)$$

Having the solution, we can examine the term $\partial A / \partial t$, which was neglected in obtaining it. By Eqs. 6, 8, 9 and B-9, we have

$$\left[\frac{\partial A}{\partial t} / q_l \right]_{x=0} = \frac{1}{16} \left[\frac{2\pi(1-\nu)\mu q_1^2}{GC^5 h} \right]^{\frac{1}{4}} t^{-\frac{3}{8}} \quad \text{. (B-10)}$$

Thus, $\partial A / \partial t$ is negligible compared with q_l at $x = 0$ (and probably over most of the fracture) for sufficiently large t . An equation of the same form as Eq. B-10 can be derived for

$$\int_0^L \frac{\partial A}{\partial t} dx / q_1 \quad .$$

APPENDIX C

NO FLUID LOSS

In this section we consider the case of no-fluid flow out of the fracture, i.e.,

$$q_l = 0 \quad \text{. (C-1)}$$

Then, substitution of Eqs. 5, 6 and C-1 into Eq. 2 gives

$$K \frac{\partial^2}{\partial x^2} (W^4) = \frac{\partial W}{\partial t} \quad \text{. (C-2)}$$

where

$$K = \frac{G}{64(1-\nu)\mu h} \quad \text{. (C-3)}$$

By Eq. 5, the boundary condition, Eq. 11 becomes

$$-\left[\frac{\partial}{\partial x} (W^4) \right]_{x=0} = \frac{4q_1}{\pi Kh} \quad \text{. (C-4)}$$

The nonlinear partial differential equation (Eq. C-2) together with Eqs. 12, 13 and C-4 could be solved by the numerical method of Appendix A.

In order to minimize computations and provide insight into the effect of various parameters, a similarity solution will be obtained. (Solutions of this type are discussed by Ames.¹²) We seek a solution of the form

$$W = \alpha \psi(\xi) t^m, \quad \xi = \beta x t^n \quad \text{. . . (C-5)}$$

where the constants α , β , m and n and the function $\psi(\xi)$ are to be determined. Substitution of Eq. C-5 into Eqs. C-2, C-4, 12 and 13 leads to the equations

$$\begin{aligned} \frac{d^2}{d\xi^2} (\psi^4) &= \frac{1}{5} \left(\psi - 4\xi \frac{d\psi}{d\xi} \right), \quad \xi > 0 \\ - \left[\frac{d}{d\xi} (\psi^4) \right]_{\xi=0} &= \frac{2}{\pi}, \quad \lim_{\xi \rightarrow \infty} \psi(\xi) = 0 \end{aligned} \quad \text{. (C-6)}$$

with

$$\begin{aligned} \alpha &= \left(\frac{4q_1^2}{h^2 K} \right)^{\frac{1}{5}}, \quad \beta = \left(\frac{h^3}{8Kq_1^3} \right)^{\frac{1}{5}} \\ m &= \frac{1}{5}, \quad n = -\frac{4}{5} \quad \text{. (C-7)} \end{aligned}$$

The function $\psi(\xi)$ can be obtained numerically by Eq. C-6. Alternately, the original problem, Eqs. C-2, C-4, 12 and 13, can be solved numerically for W , and then $\psi(\xi)$ can be obtained from Eqs. C-5 and C-7. By the latter approach, using the method of Appendix A, we find $\psi(\xi)$ and

$$\psi(\xi^*) = 0 \quad \text{at} \quad \xi^* = 1.01$$

$$\psi(0) = 0.83 \quad \text{. (C-8)}$$

Thus, by Eqs. C-3, C-5 and C-7 the fracture length is

$$L = 0.68 \left[\frac{Gq_1^3}{(1-\nu)\mu h^4} \right]^{\frac{1}{5}} t^{\frac{4}{5}} \quad \text{. (C-9)}$$

with

$$\frac{x}{L} = \frac{\xi}{\xi^*} \quad .$$

The function $\psi(\beta x t^{-4/5}) / \psi(0) = W(x, t) / W(0, t)$ is displayed in Fig. 4. By Eqs. 4, C-3, C-5 and C-7, the maximum width and pressure at the origin are

$$\begin{aligned} W(0, t) &= 2.5 \left[\frac{(1-\nu)\mu q_1^2}{Gh} \right]^{\frac{1}{5}} t^{\frac{1}{5}} \\ \Delta p(0, t) &= 2.5 \left[\frac{G^4 \mu q_1^2}{(1-\nu)^4 h^6} \right]^{\frac{1}{5}} t^{\frac{1}{5}} \end{aligned} \quad \text{. (C-10)}$$

By Eq. C-10, we have

$$\frac{\partial W}{\partial t} \sim t^{-\frac{4}{5}} \dots \dots \dots (C-13)$$

Thus, at $t \rightarrow 0$, the term $\partial W/\partial t$ dominates the $8C$ -term in Eq. 14 except near $x = L$. Therefore, the foregoing solution with $C = 0$ should be a good approximation to the solution for $C \neq 0$ at sufficiently small times.

APPENDIX D

PERKINS AND KERN APPROXIMATION

For ease of comparison, we record here the approximate solution of Perkins and Kern.⁶ Upon neglect of fluid-loss and rate of fracture volume change in the continuity equation (Eq. 2), the right-hand side of Eq. 14 is zero and integration with Eqs. 15 and 16 yields

$$W = 4 \left[\frac{(1 - \nu)\mu q_1 L}{\pi G} \left(1 - \frac{x}{L} \right) \right]^{1/4} \dots \dots \dots (D-1)$$

This equation contains fracture length, L , as an unknown. Further, the flow rate, $q(x)$, is equal to q_i everywhere including $x = L$, which is unrealistic.

For comparison, our large-time solution on elimination of C between Eqs. B-5 and B-7 gives

$$W = 4 \left[\frac{2(1 - \nu)\mu q_1 L}{\pi^2 G} \right]^{1/4} \phi \left(\frac{x}{L} \right), \dots \dots \dots (D-2)$$

where $\phi(\zeta)$ is given by Eq. B-8. Thus, Eq. D-1 differs from Eq. D-2 at $x = 0$ by the factor 0.89.

At small times, an approximate solution for L can be obtained from the over-all continuity equation obtained by integration of Eq. 14 with $C = 0$ and $\partial W/\partial t$ from Eq. D-1. In this manner, we obtain Eq. C-9 with the numerical factor replaced by 0.60. Further, a correction to Eq. D-1 can be obtained from Eq. 14 with $\partial W/\partial t$ calculated from Eq. D-1. At $x = 0$ the corrected value of W differs from Eq. D-1 by the factor 0.94. Thus, Perkins and Kern's result (Eq. D-1) for width in terms of length is a fair

approximation at early time. However, length must be determined from Eq. C-9 and not Carter's formula.²

APPENDIX E

EFFECT OF SPURT LOSS ON FRACTURE LENGTH

We replace Eq. 9 by

$$q_l = \frac{C}{\sqrt{t - \tau}} + V_s \delta(t - \tau), \dots \dots (E-1)$$

where V_s is the spurt loss (volume/area) occurring instantaneously at $t = \tau$ and $\delta(t - \tau)$ is the Dirac delta symbol. We follow the method of Appendix B and obtain instead of Eq. B-4

$$C \int_0^t \frac{dL}{d\tau} \frac{d\tau}{\sqrt{t - \tau}} + V_s \frac{dL}{dt} = \frac{q_1}{2h} \dots (E-2)$$

Eq. E-2 can be solved by the method of Laplace transform with the result

$$L = \frac{q_1}{\pi Ch} \left[t^{\frac{1}{2}} + \frac{V_s}{2C} (L b^2 t \operatorname{erfc} b\sqrt{t} - 1) \right], \dots \dots \dots (E-3)$$

where

$$b = \frac{\pi^{\frac{1}{2}} C}{V_s} \dots$$

Since

$$e b^2 t \operatorname{erfc} b\sqrt{t} = \frac{1}{\pi^{\frac{1}{2}} b} \left(1 - \frac{1}{2b^2 t} + \dots \right) t^{-\frac{1}{2}} \dots \dots \dots (E-4)$$

the V_s term is negligible in Eq. E-3 for

$$t \gg (V_s/C)^2 \dots \dots \dots (E-5)$$

which is a small time for many fracturing fluids.^{1, 11}
